

IC Lab Midterm

2021 Fall

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Total score: 115%

1. [4%] Lab01:

- (a) [2%] What is combinational loop? Use a example code to demonstrate.
- (b) [2%] Why should we prevent combinational loop in our design?

2. [6%] Lab01:

- (a) [4%] What might be the potential problem below when we want to check the answer in your PATTERN? out_valid theoretically would be 1 if output data is ready, and 0 if output data is unavailable. However, some cases indicate that out_valid would be neither 1 nor 0.

(hint: try to think about the behavior of out_valid signal, you need to describe **what situation it will lead to 0 lat value and fail to check your answer and why**. If you not mention it, you cannot get the point)

- (b) [2%] How can we solve this problem?

```
//=====
// ans task
//=====
task wait_outvalid ; begin
    lat = 0;
    while(out_valid == 0)begin
        lat = lat + 1;
        if(lat == 400) begin
            fail;
            repeat(2)@(negedge clk);
            $finish;
        end
    end
    @(negedge clk);
    end
    total_latency = total_latency + lat;
end endtask

task check_ans ; begin
    if(out_valid == 1) begin
        c = $fscanf(pat_file,"%d",&golden_match,&golden_index);
        if( (match != golden_match) || (match_index != golden_index)) begin
            fail;
            $display ("-----");
            $display ("                      FAIL!                      ");
            $display ("-----");
            @(negedge clk);
            $finish;
        end
    end
end endtask
```

```
for(p=0;p<pattern_num;p=p+1)begin
    pattern_task;
    wait_outvalid;
    check_ans;
    delay_task;
    @(negedge clk);
end
```

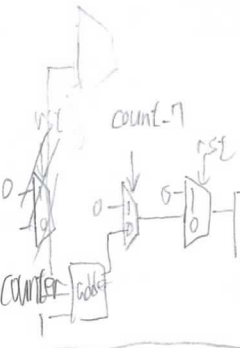
3. [5%] Lab01:

- (a) [2%] What is cell-based design methodology?
- (b) [3%] Briefly list cell-based design flow.
Front-end: ____ level design ---> ____ level design ---> ____ level design
Back-end: F_____ ---> P_____ ---> R_____

4. [15%] Lab02:

Here is a **positive edge trigger** D-Flip Flop which stores a counter. The counter will count up to 7 then return to 0.

- (a) [4%] If the D Flip-Flop with synchronous active-high reset, please fill out and finish the Verilog code below.



```

Input clk, rst;
reg [2:0] counter;
wire count_7;
assign count_7 = (counter == 'd7);
always @( ① ) begin
    if ( ② ) counter <= 0;
    else counter <= (count_7)? 'd0 : counter + 'd1;
end
    
```

- (b) [4%] Please draw the block diagram of question (a). (D-FF, Mux, adder.....)
- (c) [4%] Please list one advantage and one disadvantage of D-Flip Flop with asynchronous reset and synchronous reset.
- (d) [3%] If the Verilog code of question (a) is modified by removing line 6 as the figure shows below. Whether this counter is working normally? If not, please write down the problem that will occur.

```

Input clk, rst;
reg [2:0] counter;
wire count_7;
assign count_7 = (counter == 'd7);
always @( ① ) begin
    // if ( ② ) counter <= 0;
    else counter <= (count_7)? 'd0 : counter + 'd1;
end
    
```

5. [6%] Lab03: (Appendix Fig. 1 & Fig. 2)

- (a) [4%] What is the difference between \$display, \$write, \$strobe and \$monitor? Please fill in the table below(0.5% each blank):

	Build in Newline("\n") (O/X)	Timing (before/after)*
\$display	0	before
\$write	X	
\$strobe	0	after
\$monitor	0	after

* before/after Update LHS of nonblocking assignments

- (b) [2%] Following the previous question, what is the difference between strobe & monitor?

6. [8%] Lab03:

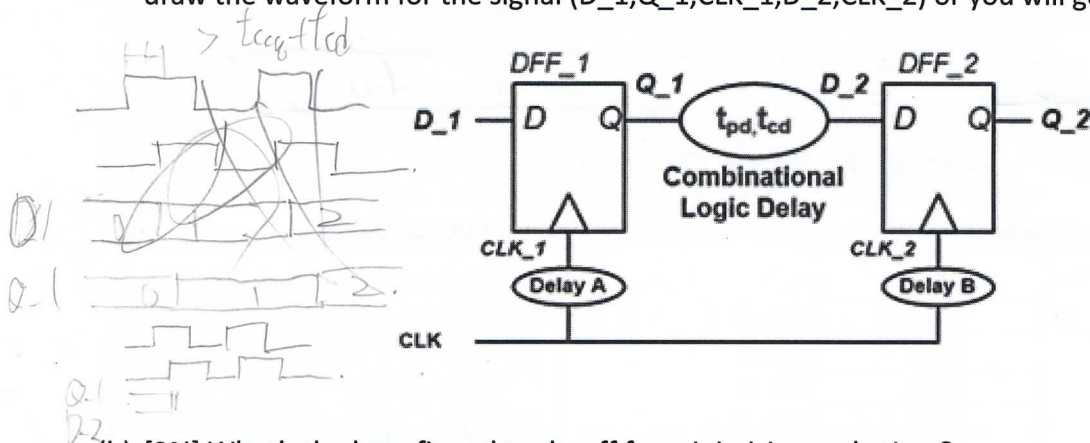
- (a) [2%] What is a direct test? *broad-spectrum*
 (b) [2%] What is a random test? *corner case*
 (c) [4%] Please give one of direct test advantage and one of its disadvantages (2%, 2%)

7. [6%] Lab03: (Appendix Fig. 3)

Please list at least three functionalities or role of TESTBED.v and PATTERN.v each (3%, 3%)

8. [16%] Lab04:

- (a) [5%+5%] Please explain two kinds of timing violation with the picture below. (You need to draw the waveform for the signal (D_1, Q_1, CLK_1, D_2, CLK_2) or you will get 0 point)



- (b) [3%] What's the benefit and trade-off for minimizing cycle time?
 (c) [3%] Please explain the reason why we need "//synopsys translate_off", "//synopsys translate_on" in our code when we use the IP.

9. [6%] Lab05:

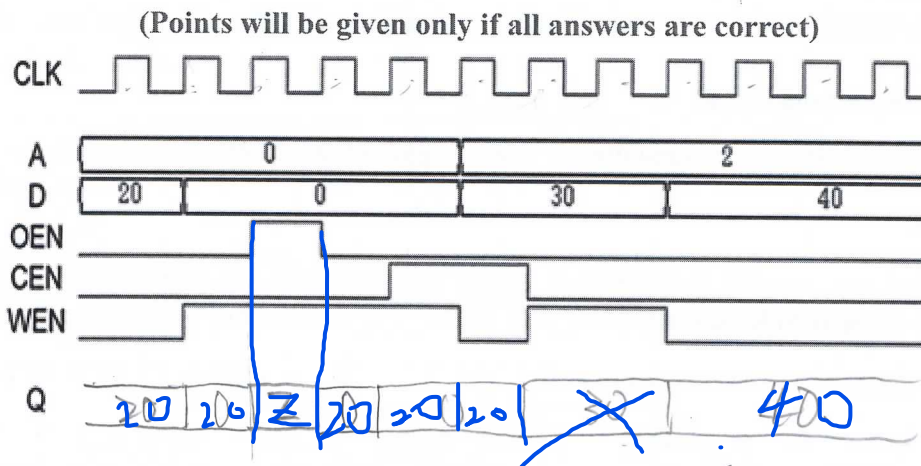
- (a) [2%] There are two memory files generated by memory compiler when we need to use memory, (*.v) and (*.lib) file, please briefly explain the (*.v) and (*.lib) file of memory.
 (b) [2%] We generate (*.db) from (*.lib), please explain why we need (*.db) file?
 (c) [2%] What's (*.sdf)'s full name and why do we need it?

10. [19%] Lab05:

- (a) [4%] The control signals of a memory are given in the table. Please fill out what're the **data outputs and modes** of (1)~(8) in the table.

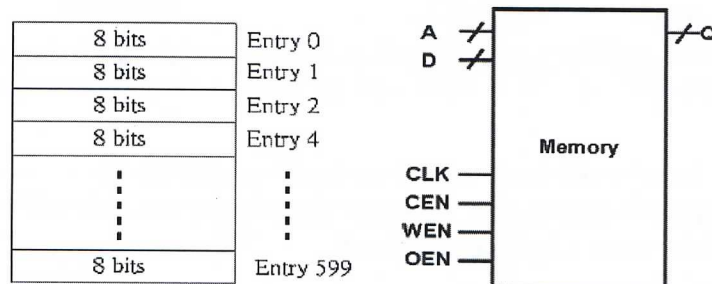
OEN	CEN	WEN	Data Output	Mode
H	X	X	<i>Z₍₁₎</i>	<i>High Z</i>
L	H	X	<i>Last Data</i>	<i>Standby</i>
L	L	H	<i>Read Data</i>	<i>Read</i>
L	L	L	<i>Write Data</i>	<i>Write</i>

(b) [4%] Please draw the waveform of the signal Q.



(c) [2%] What's the meaning of "N" in OEN, CEN, WEN? *negative*

(d) [7%] Generate a SRAM whose number of words is 600, number of bits is 8.



- [2%] How many bits of input pins (D, Q) are needed?
- [2%] How many bits of address (A) are needed?
- [3%] Which multiplexer width (4, 8, 16) should be selected for making footprint close to square? Why?

(e) [2%] Why do we need to set a "Frequency" value when generating memory?

11. [12%] Lab06:

- [3%] Write two ways to read files for synthesis.
- [3%] When to use them?
- [6%] Compare the differences between these 2 ways.

*analyze
enlabore - readverilog*

12. [12%] Lab06:

- [2%] What's the meaning of '*' when reading libraries "set link_library {*}"
- [2%] What's the meaning of '*' when selecting objects "[get_ports *]"?
- [2%] Why do we need to set input/output loading?
- [6%] What is logic synthesis? Please list 3 main steps and describe how it works for each.

Appendix

```
reg [3:0] a, b;
initial begin
    $monitor("monitor: a = %d, b = %d\n", a, b);
    a = 0;
    b = 1;
    #10;
    $display("first swap");
    a <= b;
    b <= a;
    $display("display: a = %d, b = %d\n", a, b);
    $write("write: a = %d, b = %d\n", a, b);
    $strobe("strobe: a = %d, b = %d\n", a, b);
    #10
    $display("second swap");
    a <= b;
    b <= a;
    #10
    $display("finish");
    $finish;
end
```

Fig 1: Verilog code for question 5

```
monitor: a = 0, b = 1

first swap
display: a = 0, b = 1

write: a = 0, b = 1
strobe: a = 1, b = 0

monitor: a = 1, b = 0

second swap
monitor: a = 0, b = 1

finish
```

Fig 2: Corresponding output

```
`timescale 1ns/10ps
`include "PATTERN.v"
`ifdef RTL
    `include "SMC.v"
`endif
`ifdef GATE
    `include "SMC_SYN.v"
`endif
module TESTBED;

//Connection wires
wire [2:0] W_0, V_GS_0, V_DS_0;
wire [2:0] W_1, V_GS_1, V_DS_1;
wire [2:0] W_2, V_GS_2, V_DS_2;
wire [2:0] W_3, V_GS_3, V_DS_3;
wire [2:0] W_4, V_GS_4, V_DS_4;
wire [2:0] W_5, V_GS_5, V_DS_5;
wire [1:0] mode;
wire [9:0] out_n;
initial begin
    `ifdef RTL
        $fsdbDumpfile("SMC.fsdb");
        $fsdbDumpvars(0,"+mda");
        $fsdbDumpvars();
    `endif
    `ifdef GATE
        $sdf_annotate("SMC_SYN.sdf", DUT_SMC);
        $fsdbDumpfile("SMC_SYN.fsdb");
        $fsdbDumpvars(0,"+mda");
        $fsdbDumpvars();
    `endif
end

SMC DUT_SMC(
    .W_0(W_0), .V_GS_0(V_GS_0), .V_DS_0(V_DS_0),
    .W_1(W_1), .V_GS_1(V_GS_1), .V_DS_1(V_DS_1),
    .W_2(W_2), .V_GS_2(V_GS_2), .V_DS_2(V_DS_2),
    .W_3(W_3), .V_GS_3(V_GS_3), .V_DS_3(V_DS_3),
    .W_4(W_4), .V_GS_4(V_GS_4), .V_DS_4(V_DS_4),
    .W_5(W_5), .V_GS_5(V_GS_5), .V_DS_5(V_DS_5),
    .mode(mode),
    .out_n(out_n)
);

PATTERN My_PATTERN(
    .W_0(W_0), .V_GS_0(V_GS_0), .V_DS_0(V_DS_0),
    .W_1(W_1), .V_GS_1(V_GS_1), .V_DS_1(V_DS_1),
    .W_2(W_2), .V_GS_2(V_GS_2), .V_DS_2(V_DS_2),
    .W_3(W_3), .V_GS_3(V_GS_3), .V_DS_3(V_DS_3),
    .W_4(W_4), .V_GS_4(V_GS_4), .V_DS_4(V_DS_4),
    .W_5(W_5), .V_GS_5(V_GS_5), .V_DS_5(V_DS_5),
    .mode(mode),
    .out_n(out_n)
);

endmodule
```

Fig 3: Example TESTBED